Review: Integer operations

1 Evaluate the following:

a $9 + (-7) - 4$

b $-9 + 6 - (-17)$

c $20 + (-14) - (-2) - 18$

d $6 - (13 - 3)$

e $-18 + 21 \div 7$

f $-4 \times (-7 + 18)$

g $(3 - 15) \times (-14 + 9)$

h $70 - 8 \times (-7)$

i $(-36) \div (-4) - 14$

j $(-390) \div (-19 + 6)$

k $-7 + 8 \times 6 \div 2$

l $54 \div 3 \div (-2)$

m $(-100) \div (-4) \div (-5)$

n $72 \div 6 \div 2$

o $(120 \div 5) \times 3.$

p $6 - 25 \div 5 + 4$

q $9 \times (-6) + 28 \div (-7)$

r $(-14) \div 2 - 10 \div (-2)$

s $2 \times (-25 - 20) \div (-7 + 2)$

2 Complete the following equation with $<$, $>$ or $=$ to make the statement true.

$$6 + (3 + (-11)) \square - 2 + (4 + (-6))$$

3 State the first step to simplify the following expressions:

a $-6 \div 2 \times 4$

b $-14 \div (5 - 3 \times 4)$

4 Complete the following statements to make each of the following equations true:

a $-3 \times 4 - \square = -14$

b $2 - (-3 \times \square) = 17$

c $\square + 5 \times 2 = 4$

d $(-17 - 3) \div \square = -4$

Additional questions

5 Valentina dives off a platform that is $8\ \text{m}$ above the water. She descends $14\ \text{m}$ before returning to the surface.

a If Valentina's initial position on the diving platform is represented by the integer 8, find the integer that represents her lowest position throughout the dive.

b Hence, state the greatest depth that Valentina reaches during the dive.

6 One evening in Moscow the temperature fell to -3°C. By midday the next day, the

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7 Liquid nitrogen can be used to make ice cream. The ice cream mixture starts at $73\ ^\circ\text{F}$, then rapidly cools to $-301\ ^\circ\text{F}$ after liquid nitrogen is added to it. State the integer that represents this change in temperature.

8 Three contestants are heading into the final round of a game show. Eileen has 39 points and Roxanne has -3 points. If Oprah has 14 more points than Eileen, calculate how many points separate Roxanne and Oprah.

9 A submarine is $148\ \text{m}$ below sea level. It rises $34\ \text{m}$, then dives $47\ \text{m}$. Find the distance of the submarine below sea level.

10 Each bottle of orange juice is supposed to contain $250\ \text{mL}$. Nine bottles are inspected by the quality assurance department. The results of the inspection are shown in the table:

Bottle	1	2	3	4	5	6	7	8	9	10
Amount below or above the required volume (mL)	$+4$	-7	-11	$+9$	$+3$	0	-1	-5	$+6$	$+1$

a Find the actual volume of:

i Bottle 1 **ii** Bottle 2 **iii** Bottle 6 **iv** Bottle 10

b Calculate the total volume contained in the 10 bottles.

11 Each letter has been assigned to an integer. A word must be formed using these letters and each letter can only be used once. The table shows the assignment of the integers:

Letter	A	E	I	C	T	N	R	S
Integer	-3	-2	-1	0	1	2	3	4

The score of the word is the sum of the integers assigned to its letters. Find the score of the following words:

a NET **b** EAR **c** RAIN **d** CARES

12 Are the following statements true or false? Explain.

a The product of a positive number with a negative number is negative.

b The product of an even number of negative numbers is negative.

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13 Is the quotient of a positive integer and a negative integer positive or negative? Explain.

14 Find the value of n in the following statements:

a A number n is multiplied with -3 to give 21.

b A number n is multiplied with -10 to give -40.

15 Find the value of n in the following statements:

a A number n, when divided by 3, gives -9.

b A number n when divided by -5 gives -4.

16 Which quadrant(s) could the point (x, y) lie in if:

a $xy > 0$ **b** $\dfrac{x^2}{y} < 0$

17 In golf your score for each hole is determined by comparing it to the par (or expected) number of shots for that hole. Taking one less shot than expected will give a score of -1, one more shot will give $+1$, and so on.
Ivan played 18 holes of golf, and the frequency of each score he obtained is shown in the adjacent table:

Score	Frequency
-1	5
Par	3
$+1$	4
$+2$	6

a Find Ivan's total score for his 6 scores of $+2$.

b Find Ivan's total score for his 5 scores of -1.

c Find Ivan's total score for the total 18 holes.

18 Victoria earns $\$1100$ a month, and deposits it into her bank account. She also withdraws $\$200$ a month to pay for rent.

a How much, in total, does Victoria deposit into her bank account in one full year?

b How much, in total, does Victoria withdraw from her bank account in one full year?

c Find the total change in her bank account over the year. An increase will be represented by a positive number and a decrease will be represented by a negative number.

19 Explain why the following statement is incorrect, and find the correct answer:

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20

a The temperature in Paris is $13\ ^\circ\mathrm{C}$ degrees Celsius less than double the temperature in London. If it is $8^\circ\mathrm{C}$ in London then what is the temperature in Paris?

b The temperature in Sydney is $21\ ^\circ\ \mathrm{F}$ more than triple the temperature in New York. If it is $9\ ^\circ\mathrm{F}$ in New York then what is the temperature in Sydney?

21 Laura and a group of 11 friends are ordering some pizza. They think that they will eat two pizzas for every three people. Pizzas cost $\$14$ each and there is a $\$8$ delivery fee.
If they split the total cost evenly between each person, how much will Laura need to pay?

22 At the start of the week, Quiana had $\$131$ in her bank account. On Monday, she withdrew $\$43$. On Wednesday, she deposited $\$11$, and on Thursday, she deposited three times as much as she deposited on Wednesday. Finally, she withdrew $\$25$ on Friday.
Find Quiana's final bank balance at the end of the week.

23 Starting from his house, Jerry walked $1\ \mathrm{m/s}$ due south for 30 minutes, then $2\ \mathrm{m/s}$ due north for 15 minutes and then, then due south again $2\ \mathrm{m/s}$ for 45 minutes. Find Jerry's final distance from his house.

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